

Quantum Mechanics

Complete Mastery Guide for GATE 2026

Highest Weightage Topic - 25-30%

Comprehensive Coverage with Problem-Solving Strategies

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1 Introduction

Critical for GATE

Quantum Mechanics is the **HIGHEST** weightage topic in GATE Physics, accounting for 25-30% of questions. Mastering QM is **NON-NEGOTIABLE** for a good GATE score. This guide provides complete coverage with **NO STONE LEFT UNTURNED** - every concept, every formula, every technique you need to ace this topic.

1.1 Why Quantum Mechanics is Crucial

1. **Maximum weightage:** 6-8 questions out of 25 physics questions
2. **Foundation for:** Atomic, Molecular, Nuclear, Solid State Physics
3. **Predictable patterns:** Same problem types appear regularly
4. **High scoring:** Once mastered, fastest to solve
5. **Career relevance:** Essential for research in modern physics

1.2 Study Strategy

Problem-Solving Tip

Recommended Approach:

- **Week 1-2:** Formalism (operators, commutators, eigenvalues)
- **Week 3-4:** 1D problems (wells, oscillator, barriers)
- **Week 5:** Angular momentum (spin, addition)
- **Week 6-7:** Approximation methods
- **Week 8:** Scattering theory
- Practice minimum 100 problems total!

2 Quantum Formalism: Operators and Eigenvalues

2.1 Postulates of Quantum Mechanics

Postulate 2.1 (State of a System). The state of a quantum system is completely described by a wave function $\psi(\vec{r}, t)$ satisfying:

$$\int |\psi(\vec{r}, t)|^2 d^3r = 1$$

$|\psi(\vec{r}, t)|^2$ gives the probability density.

Postulate 2.2 (Observables as Operators). Every observable physical quantity is represented by a Hermitian operator.

Postulate 2.3 (Measurement). When an observable A is measured, the only possible results are the eigenvalues of operator $\hat{A}$:

$$\hat{A}|\psi_n\rangle = a_n|\psi_n\rangle$$

Postulate 2.4 (Schrödinger Equation). Time evolution is governed by:

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{H}|\psi\rangle$$

2.2 Operators in Quantum Mechanics

Definition 2.5. An operator $\hat{A}$ acts on a state to produce another state:

$$\hat{A}|\psi\rangle = |\phi\rangle$$

Critical for GATE

Fundamental Operators (Position Representation):

$$\text{Position: } \hat{x} = x \quad (1)$$

$$\text{Momentum: } \hat{p}_x = -i\hbar \frac{\partial}{\partial x} \quad (2)$$

$$\text{Kinetic Energy: } \hat{T} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m} \nabla^2 \quad (3)$$

$$\text{Hamiltonian: } \hat{H} = \hat{T} + \hat{V} = -\frac{\hbar^2}{2m} \nabla^2 + V(\vec{r}) \quad (4)$$

Angular Momentum:

$$\vec{\hat{L}} = \vec{r} \times \hat{\vec{p}} = -i\hbar(\vec{r} \times \nabla) \quad (5)$$

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi} \quad (6)$$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 \quad (7)$$

2.3 Eigenvalues and Eigenfunctions

Definition 2.6. If $\hat{A}|\psi\rangle = a|\psi\rangle$, then:

- a is an **eigenvalue** of $\hat{A}$
- $|\psi\rangle$ is an **eigenfunction** (eigenstate) of $\hat{A}$

Theorem 2.7 (Properties of Hermitian Operators). For Hermitian operator $\hat{A}$ (representing observable):

1. All eigenvalues are real
2. Eigenfunctions belonging to different eigenvalues are orthogonal
3. Eigenfunctions form a complete set
4. $\langle\hat{A}\rangle = \langle\psi|\hat{A}|\psi\rangle$ is real

Example 2.8. Momentum Eigenfunction

Find eigenfunction of $\hat{p}_x = -i\hbar\frac{\partial}{\partial x}$.

Solution:

$$-i\hbar\frac{d\psi}{dx} = p\psi$$

$$\frac{d\psi}{dx} = \frac{ip}{\hbar}\psi$$

$$\psi(x) = Ae^{ipx/\hbar}$$

This is a plane wave with definite momentum p .

2.4 Commutators

Definition 2.9. The **commutator** of operators $\hat{A}$ and $\hat{B}$ is:

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

Critical for GATE

Fundamental Commutation Relations (Must Memorize):

$$[\hat{x}, \hat{p}_x] = i\hbar \quad (8)$$

$$[\hat{x}, \hat{p}_y] = 0 \quad (9)$$

$$[\hat{x}, \hat{x}] = 0, \quad [\hat{p}_x, \hat{p}_x] = 0 \quad (10)$$

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k \quad (11)$$

$$[\hat{L}^2, \hat{L}_i] = 0 \quad (\text{for any component}) \quad (12)$$

Angular Momentum Commutators:

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z \quad (13)$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x \quad (14)$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y \quad (15)$$

Theorem 2.10 (Simultaneous Observables). Two observables can be measured simultaneously with arbitrary precision if and only if their operators commute:

$$[\hat{A}, \hat{B}] = 0$$

In this case, they share a common set of eigenfunctions.

Theorem 2.11 (Uncertainty Principle). For any two observables A and B :

$$\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$$

For position and momentum:

$$\Delta x \cdot \Delta p_x \geq \frac{\hbar}{2}$$

For energy and time:

$$\Delta E \cdot \Delta t \geq \frac{\hbar}{2}$$

2.5 Properties of Commutators

Formula 2.12 (Commutator Algebra).

$$[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}] \quad (\text{antisymmetric}) \quad (16)$$

$$[\hat{A}, \hat{B} + \hat{C}] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}] \quad (\text{linear}) \quad (17)$$

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}] \quad (\text{Leibniz}) \quad (18)$$

$$[\hat{A}\hat{B}, \hat{C}] = \hat{A}[\hat{B}, \hat{C}] + [\hat{A}, \hat{C}]\hat{B} \quad (19)$$

$$[\hat{A}, [\hat{B}, \hat{C}]] + [\hat{B}, [\hat{C}, \hat{A}]] + [\hat{C}, [\hat{A}, \hat{B}]] = 0 \quad (\text{Jacobi}) \quad (20)$$

Example 2.13. Calculate $[\hat{x}^2, \hat{p}_x]$

Solution:

$$\begin{aligned} [\hat{x}^2, \hat{p}_x] &= \hat{x}[\hat{x}, \hat{p}_x] + [\hat{x}, \hat{p}_x]\hat{x} \\ &= \hat{x}(i\hbar) + (i\hbar)\hat{x} \\ &= 2i\hbar\hat{x} \end{aligned}$$

2.6 Expectation Values

Definition 2.14. The **expectation value** of operator $\hat{A}$ in state $|\psi\rangle$ is:

$$\langle A \rangle = \langle \psi | \hat{A} | \psi \rangle = \int \psi^* \hat{A} \psi d^3r$$

Uncertainty:

$$\Delta A = \sqrt{\langle A^2 \rangle - \langle A \rangle^2}$$

2.7 Time Evolution

Formula 2.15 (Ehrenfest Theorem).

$$\frac{d\langle \hat{A} \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle + \left\langle \frac{\partial \hat{A}}{\partial t} \right\rangle$$

For time-independent operators:

$$\frac{d\langle \hat{A} \rangle}{dt} = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle$$

If $[\hat{A}, \hat{H}] = 0$, then $\langle A \rangle$ is conserved.

2.8 Dirac Notation and Bra-Ket Formalism

Definition 2.16. Ket: $|\psi\rangle$ represents a state vector

Bra: $\langle\psi|$ represents the dual vector (complex conjugate transpose)

Inner product: $\langle\phi|\psi\rangle = \int \phi^* \psi d^3r$

Outer product: $|\psi\rangle\langle\phi|$ is an operator

Completeness: $\sum_n |\psi_n\rangle\langle\psi_n| = \hat{I}$ (identity)

2.9 Practice Problems - Operators and Commutators

1. Verify $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$ using definitions.
2. Calculate $[\hat{x}, \hat{p}_x^2]$ and $[\hat{x}^3, \hat{p}_x]$.
3. Show that $[\hat{H}, \hat{p}_x] = 0$ for free particle.
4. Prove that if $[\hat{A}, \hat{B}] = c$ (constant), then $\Delta A \cdot \Delta B \geq |c|/2$.
5. Find commutator $[\hat{x}\hat{p}_x, \hat{L}_z]$.

3 One-Dimensional Quantum Systems

3.1 Time-Independent Schrödinger Equation

Formula 3.1. For stationary states $\psi(x, t) = \psi(x)e^{-iEt/\hbar}$:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi$$

Or:

$$\frac{d^2\psi}{dx^2} + \frac{2m}{\hbar^2}[E - V(x)]\psi = 0$$

Critical for GATE

General Strategy for 1D Problems:

1. Identify regions with different potentials
2. Write general solution in each region
3. Apply boundary conditions:
 - ψ continuous everywhere
 - ψ' continuous except at infinite potential jumps
 - $\psi \rightarrow 0$ as $x \rightarrow \pm\infty$ for bound states
4. Match solutions at boundaries
5. Apply normalization
6. Determine energy eigenvalues and eigenfunctions

3.2 Infinite Square Well (Particle in a Box)

Definition 3.2. Potential:

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & \text{otherwise} \end{cases}$$

Theorem 3.3 (Solutions for Infinite Well). **Energy eigenvalues:**

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2} = \frac{n^2h^2}{8mL^2}, \quad n = 1, 2, 3, \dots$$

Eigenfunctions:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

Properties:

- Ground state: $n = 1$, $E_1 = \frac{\pi^2\hbar^2}{2mL^2}$
- Energy spacing: $E_n \propto n^2$ (not equally spaced!)

- Orthonormal: $\int_0^L \psi_m^* \psi_n dx = \delta_{mn}$
- Zero-point energy: $E_1 \neq 0$ (quantum mechanical)

Example 3.4. Expectation Values in Infinite Well

For particle in ground state ($n = 1$):

Position: $\langle x \rangle = \int_0^L x |\psi_1|^2 dx = \frac{L}{2}$ (center of well)

Momentum: $\langle p \rangle = 0$ (symmetric probability distribution)

Energy: $\langle E \rangle = E_1 = \frac{\pi^2 \hbar^2}{2mL^2}$

3.3 Finite Square Well

Definition 3.5. Potential:

$$V(x) = \begin{cases} -V_0 & -a < x < a \\ 0 & |x| > a \end{cases}$$

where $V_0 > 0$.

Theorem 3.6 (Bound States of Finite Well). For bound states ($-V_0 < E < 0$), define:

$$k = \frac{\sqrt{2m(E + V_0)}}{\hbar}, \quad \kappa = \frac{\sqrt{-2mE}}{\hbar}$$

Solutions are either:

Even states:

$$\psi(x) = \begin{cases} A \cos(kx) & |x| < a \\ B e^{-\kappa|x|} & |x| > a \end{cases}$$

Quantization: $k \tan(ka) = \kappa$

Odd states:

$$\psi(x) = \begin{cases} A \sin(kx) & |x| < a \\ B \operatorname{sgn}(x) e^{-\kappa|x|} & |x| > a \end{cases}$$

Quantization: $k \cot(ka) = -\kappa$

Number of bound states: At least one; maximum approximately $\frac{2a}{\pi} \sqrt{2mV_0}/\hbar$

Problem-Solving Tip

Key Points for Finite Well:

- Always has at least one bound state
- Wave function penetrates into classically forbidden region
- Transcendental equations for energy levels
- Symmetric potential gives definite parity states

3.4 Quantum Harmonic Oscillator

Definition 3.7. Potential: $V(x) = \frac{1}{2}m\omega^2 x^2$

Hamiltonian: $\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 \hat{x}^2$

3.4.1 Ladder Operator Method

Critical for GATE

Creation and Annihilation Operators:

$$\hat{a} = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i\hat{p}}{m\omega} \right)$$

$$\hat{a}^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i\hat{p}}{m\omega} \right)$$

Key Relations:

$$[\hat{a}, \hat{a}^\dagger] = 1 \quad (21)$$

$$\hat{H} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) = \hbar\omega \left(\hat{N} + \frac{1}{2} \right) \quad (22)$$

Action on States:

$$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle \quad (23)$$

$$\hat{a}^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \quad (24)$$

Theorem 3.8 (Harmonic Oscillator Solutions). **Energy eigenvalues:**

$$E_n = \hbar\omega \left(n + \frac{1}{2} \right), \quad n = 0, 1, 2, \dots$$

Properties:

- Equally spaced levels (spacing $\hbar\omega$)
- Zero-point energy: $E_0 = \frac{\hbar\omega}{2}$
- Ground state: $\hat{a}|0\rangle = 0$

3.5 Potential Barrier (Tunneling)

Theorem 3.9 (Quantum Tunneling). For $E < V_0$, particle can tunnel through barrier!

Transmission coefficient:

$$T \approx e^{-2\kappa a} \quad \text{where } \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}$$

3.6 Practice Problems - 1D Systems

1. Particle in infinite well: Find $\langle x^2 \rangle$ for state n .
2. Harmonic oscillator: Calculate $\langle x \rangle$ and $\langle p \rangle$ for state $|n\rangle$.
3. Find $\langle n|\hat{x}^2|n\rangle$ using ladder operators.
4. Potential barrier: Find transmission probability for $E = V_0/2$.

4 Angular Momentum in Quantum Mechanics

4.1 Orbital Angular Momentum

Definition 4.1. Quantum angular momentum operator:

$$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}} = -i\hbar(\vec{r} \times \nabla)$$

4.2 Commutation Relations

Critical for GATE

Fundamental Angular Momentum Commutators:

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z \quad (25)$$

$$[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x \quad (26)$$

$$[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y \quad (27)$$

$$[\hat{L}^2, \hat{L}_i] = 0 \quad \text{for } i = x, y, z \quad (28)$$

4.3 Eigenvalues of Angular Momentum

Theorem 4.2 (Angular Momentum Eigenvalues).

$$\hat{L}^2|l, m\rangle = \hbar^2 l(l+1)|l, m\rangle$$

$$\hat{L}_z|l, m\rangle = \hbar m|l, m\rangle$$

where:

- $l = 0, 1, 2, 3, \dots$
- $m = -l, -l+1, \dots, l-1, l$

Magnitude:

$$|\vec{L}| = \hbar\sqrt{l(l+1)}$$

4.4 Ladder Operators for Angular Momentum

Definition 4.3. Raising and lowering operators:

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y$$

Critical for GATE

Action on states:

$$\hat{L}_{\pm}|l, m\rangle = \hbar\sqrt{l(l+1) - m(m \pm 1)}|l, m \pm 1\rangle$$

4.5 Spin Angular Momentum

Definition 4.4. For spin-1/2 particles:

$$\hat{\vec{S}} = \frac{\hbar}{2} \vec{\sigma}$$

where $\vec{\sigma}$ are Pauli matrices.

Critical for GATE

Pauli Matrices:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

4.6 Addition of Angular Momenta

Theorem 4.5 (Clebsch-Gordan Series). When combining $\vec{J}_1$ and $\vec{J}_2$:
Possible total j :

$$j = |j_1 - j_2|, |j_1 - j_2| + 1, \dots, j_1 + j_2 - 1, j_1 + j_2$$

4.7 Practice Problems - Angular Momentum

1. Verify $[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z$ using definitions.
2. Calculate $\langle 1, 0 | \hat{L}_x^2 | 1, 0 \rangle$ using ladder operators.
3. Find eigenvalues of $\hat{S}_x$ for spin-1/2.
4. Two spin-1 particles: Find possible total j .

5 Approximation Methods

5.1 Time-Independent Perturbation Theory

Critical for GATE

Energy Corrections:

First order:

$$E_n^{(1)} = \langle n^{(0)} | \hat{H}' | n^{(0)} \rangle$$

Second order:

$$E_n^{(2)} = \sum_{k \neq n} \frac{|\langle k^{(0)} | \hat{H}' | n^{(0)} \rangle|^2}{E_n^{(0)} - E_k^{(0)}}$$

5.2 Variational Method

Theorem 5.1 (Variational Principle). For any normalized trial wave function:

$$\langle \psi_{\text{trial}} | \hat{H} | \psi_{\text{trial}} \rangle \geq E_0$$

5.3 WKB Approximation

Theorem 5.2 (WKB Quantization).

$$\oint p \, dx = 2\pi\hbar \left(n + \frac{1}{2} \right)$$

5.4 Practice Problems - Approximation Methods

1. Harmonic oscillator with $\hat{H}' = \lambda \hat{x}^3$: Find $E_0^{(1)}$.
2. Use variational method for ground state of hydrogen.
3. WKB: Find energy levels for $V(x) = \alpha|x|$.

6 Scattering Theory

6.1 Scattering Cross Section

Definition 6.1. Differential cross section:

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2$$

Total cross section:

$$\sigma_{\text{tot}} = \int |f(\theta)|^2 d\Omega$$

6.2 Partial Wave Analysis

Theorem 6.2 (Partial Wave Scattering).

$$f(\theta) = \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) e^{i\delta_l} \sin \delta_l P_l(\cos \theta)$$

Total cross section:

$$\sigma_{\text{tot}} = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l$$

6.3 Born Approximation

Definition 6.3. First Born approximation:

$$f(\theta) = -\frac{m}{2\pi\hbar^2} \int e^{i\vec{q}\cdot\vec{r}} V(\vec{r}) d^3r$$

6.4 Practice Problems - Scattering

1. Show optical theorem: $\sigma_{\text{tot}} = (4\pi/k) \text{Im}[f(0)]$.
2. Born approximation for $V(r) = V_0 e^{-r^2/a^2}$.

7 GATE Exam Strategy for Quantum Mechanics

7.1 Essential Formulas

Critical for GATE

Quick Reference:

$$\begin{aligned}[\hat{x}, \hat{p}_x] &= i\hbar \\ E_n^{\text{well}} &= \frac{n^2 \pi^2 \hbar^2}{2mL^2} \\ E_n^{\text{HO}} &= \hbar\omega(n + 1/2) \\ \hat{L}^2 |l, m\rangle &= \hbar^2 l(l+1) |l, m\rangle \\ E_n^{(1)} &= \langle n | \hat{H}' | n \rangle\end{aligned}$$

7.2 Common Mistakes to Avoid

1. Forgetting $\hbar$ in commutators
2. Wrong normalization
3. Confusing $l(l+1)$ with l^2
4. Forgetting $\sqrt{n}$ in ladder operators

7.3 Last 30 Days Plan

Critical for GATE

Week 1-2: Operators and 1D systems (70 problems) **Week 3:** Angular momentum and approximations (50 problems) **Week 4:** Integration and GATE questions (50 problems)

Conclusion

Quantum Mechanics: Key to GATE Success

With 25-30% weightage, mastering QM is essential!

All the Best for GATE 2026!